

Digital Clinic Management System

Deviations from the initial proposal

- **Data source changes:** None
- **ER Diagram/Schema/Database implementation changes:** None
- **Functionalities added:** None
- **Functionalities removed:** Generation of medical/referral forms for patients

What has the application achieved or failed?

Achievements:

- **Comprehensive Appointment Management:** Doctors can create, schedule, track, and update appointments seamlessly.
- **Digital Patient Records and Profiles:** Patient profiles consolidate appointment history, vitals trends, prescriptions and invoices.
- **Organizational hierarchy and RBAC:** Admins can get a detailed view of the performance of their clinics and activity of other doctors and shared patient access within their clinic.
- **Digitized Medical Documents:** Automatic PDF generation for prescriptions and invoices. Saves time, and minimizes human errors in document preparation.

Limitations:

- **Limited Patient-Side Interaction:** While doctors have robust tools, the system does not mention patient-facing features.
- **Customizability and Flexibility:** Predefined templates for prescriptions and invoices are useful, but doctors might need highly customizable prescriptions or invoices.
- **Limited automations/suggestions:** More advanced decision support (e.g., suggesting diagnoses, flagging abnormal vitals) is not present.

Role of Advanced Database Programs in the Application

- **Group By and Aggregations:** Group data and compute values for the clinic income and patient visit trend graphs over each month/year.
- **Stored Procedures:** Set of functionality to calculate and create invoice combining data from the consultation procedures (ConsultationProcedures table) and the doctor fees (Doctors table) for each consultation/appointment.
- **Triggers:** Execute actions automatically whenever an appointment is marked complete or completed, one of which involves calling the procedure to generate the invoice.
- **Constraints:** Serves as the final layer of validation and protection over the server-side validation. Also allows to cascade changes made to Patients -> Appointments, Appointments -> Invoices, Appointments -> Consultations, etc.

- **Transactions:** Used for APIs where multiple queries are executed to complete the output, in order to maintain atomicity. Creation of doctors and clinics, consultation and consultation procedures, etc.

Technical Challenges Encountered by Team Members

Arjun - Deploying the PDF generation feature to Google Cloud Run. The feature requires an installation of Chrome on the VM, which was not possible directly using NPM due to restrictions of GCP. It involved creating a separate Docker image on top of the Node LTS image, containing the libraries required to run Chrome and Puppeteer and running both instances simultaneously. Another challenge was to set up the service accounts with the right permissions for and integrate it with IAM for Firebase and CloudSQL, to be able to use the same for all the operations.

Jahnvi - Building the analytics dashboards for clinic revenue and patient visit trends. One technical challenge was managing multiple asynchronous API calls while keeping the dashboard responsive with proper loading and empty states. Using external charting libraries was trickier than expected, so custom horizontal bar visualizations were created using CSS gradients and percentage-based width calculations. Role-based access control was also enforced so that only admin users could view the analytics.

Shiv- One technical challenge I faced was getting the patient vitals trend graph to work seamlessly with the UI while ensuring proper backend integration. Since this was my first time using Chart.js, I had to learn how its datasets, labels, and configuration options worked while simultaneously aligning them with our API structure. Handling null values was another major issue, as missing vitals caused the chart to break or render incorrectly, so I added strict null checks and fallback handling to stabilize the visualization. I also had to refine how the dates and vitals were mapped to ensure the graph reflected accurate trends.

Grace - One technical challenge for me is definitely the integration, for example, I know how to write the handles for API and sections presented in the frontend, but I found it very difficult to connect the wires, especially some of my functionality depended on other teammate's section of works.

Future Work and System Improvements

The current system provides a strong foundation for digital clinic management, but a few natural extensions could further enhance its usefulness in future iterations:

- **Notifications:** Notify patients to securely view their prescriptions, invoices, and appointment details, improving transparency and engagement.
- **Customizable Prescriptions and Invoices:** Add greater flexibility in template layouts and optional fields, so doctors can tailor documents to their clinical needs.
- **Rule-Based Vitals Checks:** Introduce simple threshold-based alerts for abnormal vitals readings to assist in basic health monitoring.

- **Suggestions for fields:** Pull up suggestions from a medical database which are related to the other fields entered during the consultation.
- **Additional consultation fields:** Fields such as medications, advice, or option to add custom fields.
- **Consultation templates:** Support for creating and applying diagnosis/medication from a previous consultation or a template.
- **Partial migration to different databases:** Move some of the unstructured data of the consultation table such as procedures/medication to a NoSQL format which is better suited for data without a defined size/structure. Suggestions can be drawn from ElasticSearch.
- **Receptionist-access:** Adding a role that can function on behalf of the admin doctor, having greater privileges than consultants, but cannot access consultation records.

These enhancements could be built naturally on the existing foundation without altering the core design of the application.

Final Division of Labor and Teamwork Evaluation

Team Member	Tasks
Arjun Sivaraman	- DDL for doctors (users) table - DDL for clinic table - APIs and CRUD for doctors - Set up application-wide authentication and role-based access - Template PDF generation - Appointment-based triggers - Invoice-based triggers - Patient search functionality for the dashboard - UI for patient profile invoices - GCP Cloud SQL and Cloud Run setup and deployment
Jahnavi Juluri	- UI for Login and Signup pages - UI for Doctor Dashboard - UI for Patient list and profile - DDL for Appointments table - APIs and CRUD for appointments - Add and delete appointment functionality - Clinic Income and Patient Visits trend graphs
Shiv Bhavsar	- DDL for Patients table - APIs and CRUD for patients - Add and delete patient functionality - Patient vitals trend graph - DDL for Procedures table - Fetch procedures API
Grace Lin	- DDL for Invoice table - APIs and CRUD for invoices - DDL for Consultation table

	- APIs and CRUD for consultation - CRUD for consultation procedures - UI for consultation screen
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